

Q1. Load the TCS_history.csv

```
import pandas as pd
import numpy as np
file_path = r"C:\Users\Asus\Desktop\TCS_stock_history.xlsx"
df = pd.read_excel(file_path)
print(df.head())
```

	Date	Open	High	Low	Close	Volume
Dividends \						
0	2002-08-12	28.794172	29.742206	28.794172	29.519140	212976
0.0						
1	2002-08-13	29.556316	30.030333	28.905705	29.119476	153576
0.0						
2	2002-08-14	29.184536	29.184536	26.563503	27.111877	822776
0.0						
3	2002-08-15	27.111877	27.111877	27.111877	27.111877	0
0.0						
4	2002-08-16	26.972458	28.255089	26.582090	27.046812	811856
0.0						
Stock Splits						
0		0				
1		0				
2		0				
3		0				
4		0				

Q2. Do the some descriptive analysis

```
print("\nSummary statistics of the DataFrame:")
print(df.describe())
```

Summary statistics of the DataFrame:				
	Date	Open	High	
Low \				
count	4463	4463.000000	4463.000000	
4463.000000				
mean	2012-08-23 19:22:31.109119488	866.936239	876.675013	
856.653850				
min	2002-08-12 00:00:00	24.146938	27.102587	
24.146938				
25%	2008-02-14 12:00:00	188.951782	191.571816	
185.979417				
50%	2012-09-04 00:00:00	530.907530	534.751639	
525.616849				
75%	2017-03-22 12:00:00	1156.462421	1165.815854	
1143.622800				

max	2021-09-30 00:00:00	3930.000000	3981.750000
3892.100098			
std	NaN	829.905368	838.267104
821.233477			

	Close	Volume	Dividends	Stock Splits
count	4463.000000	4.463000e+03	4463.000000	4463.000000
mean	866.537398	3.537876e+06	0.071533	0.001344
min	26.377609	0.000000e+00	0.000000	0.000000
25%	188.594620	1.860959e+06	0.000000	0.000000
50%	529.713257	2.757742e+06	0.000000	0.000000
75%	1154.784851	4.278625e+06	0.000000	0.000000
max	3954.550049	8.806715e+07	40.000000	2.000000
std	829.611313	3.273531e+06	0.965401	0.051842

Q3. Apply Linear Regression, polynomial, and RBF for each model find the confidence level:
confidence level= right predicted values/test size

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
X = df[['Open', 'High', 'Low', 'Volume']]
y = df['Close']
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
confidence_level_lr = model.score(X_test, y_test)
print(f'Confidence Level with Linear Regression:
{confidence_level_lr}')
```

Confidence Level with Linear Regression: 0.9999463328387083

```
from sklearn.svm import SVR
from sklearn.preprocessing import StandardScaler
scaler_X = StandardScaler()
scaler_y = StandardScaler()
X_scaled = scaler_X.fit_transform(X)
y_scaled = scaler_y.fit_transform(y.values.reshape(-1, 1)).flatten()
X_train, X_test, y_train, y_test = train_test_split(X_scaled,
y_scaled, test_size=0.2, random_state=42)
model_poly = SVR(kernel='poly', degree=2)
model_poly.fit(X_train, y_train)
y_pred_poly = model_poly.predict(X_test)
confidence_level_poly = model_poly.score(X_test, y_test)
print(f'Confidence Level with Polynomial SVR:
{confidence_level_poly}')
```

Confidence Level with Polynomial SVR: 0.4926342655781333

```

model_rbf = SVR(kernel='rbf')
model_rbf.fit(X_train, y_train)
y_pred_rbf = model_rbf.predict(X_test)
confidence_level_rbf = model_rbf.score(X_test, y_test)
print(f'Confidence Level with RBF SVR: {confidence_level_rbf}')

```

Confidence Level with RBF SVR: 0.9912544976905163

Q3. Predict the close price with random dates as input with the model with highest confidence.

```

random_dates = pd.DataFrame({
    'Date': ['2024-08-01', '2024-08-02', '2024-08-03'],
    'Open': [3500, 3550, 3600],
    'High': [3550, 3600, 3650],
    'Low': [3450, 3500, 3550],
    'Volume': [1000000, 1100000, 1200000]
})
X_random = random_dates[['Open', 'High', 'Low', 'Volume']]
predicted_close_prices = model.predict(X_random)
random_dates['Close Price'] = predicted_close_prices

print(random_dates[['Date', 'Close Price']])

```

	Date	Close Price
0	2024-08-01	3500.874146
1	2024-08-02	3550.806769
2	2024-08-03	3600.739392